# Agentic RAG for University Curriculum Navigation

## Combining Semantic Retrieval with Structured Lookup Tools

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# Outline


1. Problem & Motivation
2. System Architecture
3. Evaluation
4. Discussion & Lessons Learned
5. Demo & Future Work

# The Problem

Students need answers to questions like:

- What are the prerequisites for Deep Learning?
- Can I accelerate to reach DPL302m by semester 4?
- Which specialization combo should I choose?
- I failed MAS291 — what courses are blocked?
- How many tests does MAD101 have?

**Current situation:**

- Human advisors are overloaded
- Curriculum documents are scattered across files
- LLMs alone **hallucinate** course details

## LLM Hallucination Example

**Q:** How many credits is DPL302m?

**Baseline LLM:** "DPL302m is typically a 3-credit course. . . "

**Ground truth: 6 credits**

# Our Approach: Agentic RAG

## Key Idea

Combine **semantic vector search** (for open-ended content questions) with **deterministic structured tools** (for factual lookups), orchestrated by an **agentic loop** that calls multiple tools before answering.

**Why not plain RAG?**

- Prerequisite chains need recursive resolution
- Semester-course mappings are tabular
- Credit counts need exact lookup, not similarity

**Why not Graph RAG?**

- Only 9 source documents, 47 courses
- Relationships are explicitly listed in tables
- JSON lookup is deterministic and verifiable

# Data Corpus & Ingestion Pipeline

**Raw data:** 9 markdown files

- Syllabi (course content, assessments)
- Curriculum (47 courses, 10 semesters)
- Pathways (specialization combos)

**Ingestion produces two outputs:**

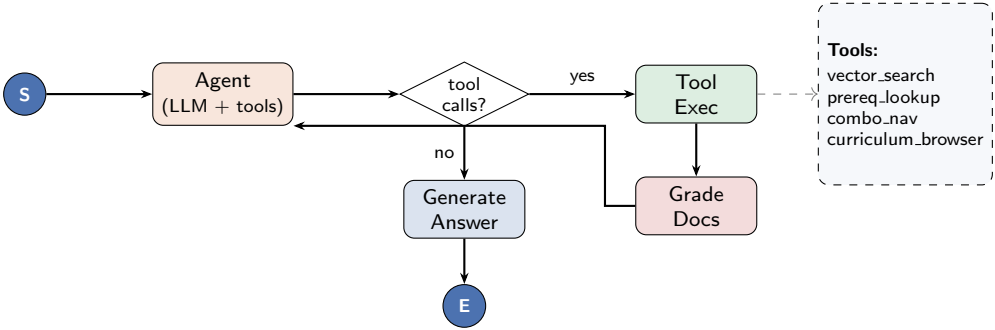
## Vector Index

- MarkdownHeaderTextSplitter (H2 sections)
- RecursiveCharacterTextSplitter (500 chars)
- Min chunk filter: 50 chars
- 8,446 → **6,256 chunks**
- OpenAI text-embedding-3-small (1536-d)
- Pinecone Serverless (cosine)

## Structured Tables

- course_index (47 courses)
- prerequisites (23 entries)
- curriculum_map (10 semesters)
- combo_map (5 pathways)
- programme_outcomes, PLOs

# Agent Graph Architecture




Agent loops up to **6 tool calls**. Structured tools skip grading. Only `vector_search` is graded (1 rewrite attempt).

# Tool Definitions

### `vector_search(q, code, type)`

Semantic search over Pinecone
*Use for:* course content, learning outcomes, assessments, session topics, materials

### `combo_navigator(id, topic)`

Browse specialization pathways
*Use for:* which combos exist, what courses in a combo, elective options

### `prerequisite_lookup(code, dir)`

Recursive prerequisite chain resolver
*Forward:* what do I need before X?
*Reverse:* what does X unlock?

### `curriculum_browser(sem, code)`

Semester-level curriculum data
*Use for:* courses in semester X, credit counts, full study plan overview

All tools include **fuzzy course code resolution**: "AIL" $\rightarrow$ "AIL303m"

# Student Personalization

**Student profiles contain:**

- Current semester
- Passed courses with grades
- Failed courses (must retake)
- Computed GPA

**FPT academic rules encoded:**

- 10-week semesters, 1-month breaks
- Max 2 extra courses/semester
- Retake allowed for GPA improvement

## Example: Struggling Student

**Pham Duc Huy** — Semester 4
**Failed:** MAS291 (3.5/10)
**Low grades:** MAE101 (5.0), MAD101 (4.5)

**Q:** Do I have to retake any course?

**Agent:** "You must retake **MAS291**. This blocks **AIL303m** (Machine Learning) and transitively **DPL302m** (Deep Learning). You can take AI17_COM*1, AI17_COM*2, and DWP301c in Semester 5."

# Experimental Setup

## Golden Test Set

**48 questions** across 5 categories:

| Category | Count |
|---|---|
| Factual Lookup | 10 |
| Prerequisite Reasoning | 10 |
| Combo Pathway | 8 |
| Cross Document | 10 |
| Curriculum Planning | 10 |

Each with reference answer and expected tool.

## Evaluation Metrics (LLM-as-Judge)

GPT-4o-mini scores on $[0, 1]$:

- **Relevancy**: addresses the question?
- **Faithfulness**: grounded in context?
- **Correctness**: matches reference?

## Systems Compared

- **RAG**: Full agent + 4 tools
- **Baseline**: GPT-4o-mini only

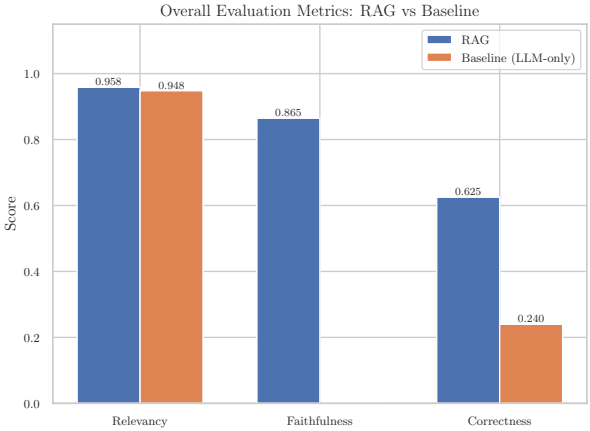
Both at temperature = 0.

# Overall Results

| Metric | RAG | Base | Δ |
|---|---|---|---|
| Relevancy | 0.958 | 0.948 | +0.010 |
| Faithfulness | **0.865** | 0.000 | **+0.865** |
| Correctness | **0.625** | 0.240 | **+0.385** |

Key findings:

- **+160.4%** relative correctness improvement
- Faithfulness: 0 → 0.865
- Relevancy comparable (both ∼0.95)



# Per-Category Correctness

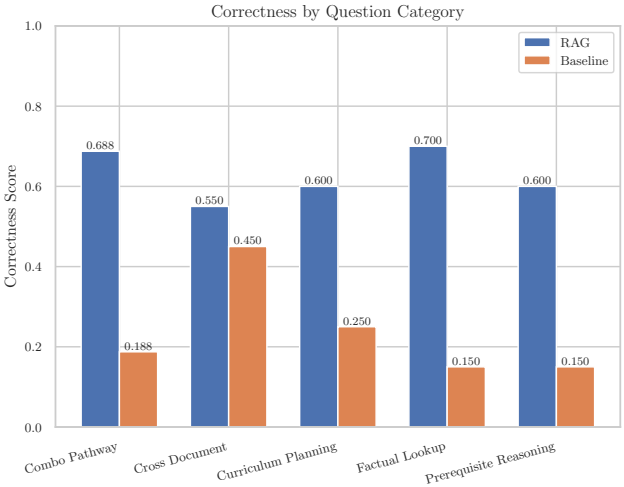
| Category | RAG | Base | Δ |
|---|---|---|---|
| Factual | 0.700 | 0.150 | +0.550 |
| Combo | 0.688 | 0.188 | +0.500 |
| Prereq | 0.600 | 0.150 | +0.450 |
| Planning | 0.600 | 0.250 | +0.350 |
| Cross-doc | 0.550 | 0.450 | +0.100 |

**Largest gains:**

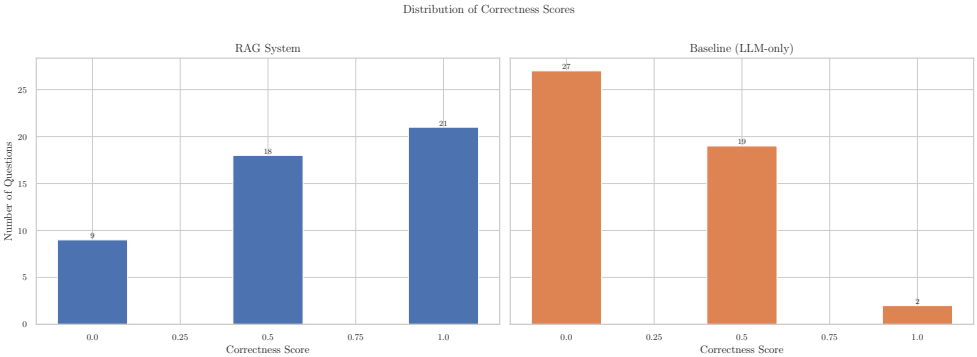
- Factual lookup: **+0.550**
- Combo pathways: **+0.500**

**Smallest gain:**

- Cross-document: +0.100

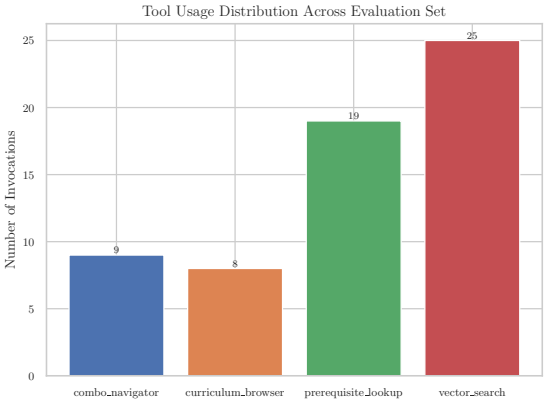


# Score Distribution




RAG produces **19 fully correct** answers (1.0) vs. **5** for baseline.
RAG has **9 failures** (0.0) vs. **27** for baseline.

# Tool Usage Patterns

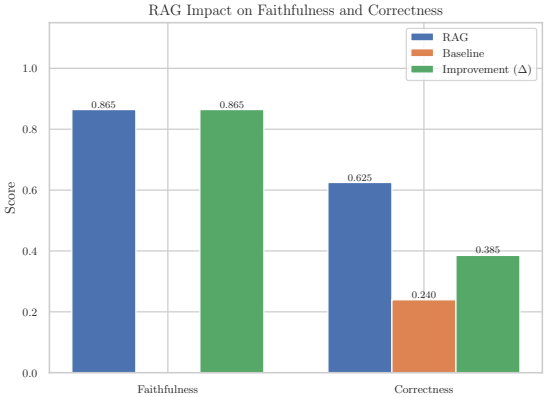



**Observations:**

- `vector_search` most frequent — content questions dominate
- `prerequisite_lookup` for dependency chains
- `curriculum_browser` for semester/credit queries
- `combo_navigator` for specialization questions

**Multi-tool queries:**
Complex questions invoke 2–3 tools in sequence.
Example: "Can I accelerate to DPL302m?"
→ `prerequisite_lookup` + `curriculum_browser`

# Why Faithfulness Matters




**Baseline (no tools):**

- Faithfulness = **0.000**
- No context → no grounding
- Invents course codes, credits, semesters

**RAG (with tools):**

- Faithfulness = **0.865**
- Answers cite actual tool results
- Verifiable against source data

**Takeaway**

Tool access is *necessary* for trustworthy curriculum advising. LLMs alone cannot be trusted with factual academic data.

# Failure Analysis

**RAG failures** (9/48 = 18.8%):

1. **Credit confusion** (fact_01, 04, 09)
   vector_search retrieves chunks without credit data; curriculum_browser not called as fallback
2. **Incomplete prereq chains** (prereq_03, 06)
   Deep 3+ level hierarchies partially resolved
3. **Cross-doc synthesis** (content_03, 09)
   Multi-source queries need multiple vector searches; coverage gaps

**Baseline failures** (27/48 = 56.3%):

**Systematic Errors**

- Invents nonexistent course codes
- Wrong semester numbers (sem 6 instead of 4)
- Inaccurate totals (40 subjects vs. actual 47)
- Fabricates prerequisite relationships

**Key difference:**
RAG failures are *incomplete* answers.
Baseline failures are *incorrect* answers.

# Key Design Decisions

1. **Structured tools over Graph RAG**
   - 9 documents, 47 courses — too small for knowledge graph overhead
   - All relationships explicitly listed → JSON lookup is deterministic
   - Result: structured tools achieve higher correctness than vector search
2. **Multi-tool loop instead of single-pass**
   - Original design: agent → tool → generate (one tool only)
   - Problem: "Can I accelerate to DPL302m?" needs 2+ tools
   - Fix: agent ↔ tools loop with safety counter (max 6)
3. **Current-turn-only generation**
   - Problem: generate node saw full conversation history, repeated old answers
   - Fix: extract only current question + its tool results for synthesis
4. **Selective grading**
   - Only `vector_search` graded — structured tools are always relevant
   - Query rewriting preserves course code prefixes (avoids "MAE" → "Master of Arts")

# Live Demo

*Streamlit UI Demo*

| Feature | Description |
|---|---|
| Profile selector | 5 predefined student personas |
| Chat interface | Multi-turn conversation |
| Source citations | Expandable tool call results |
| Profile details | Passed/failed courses, GPA |

```
$ streamlit run src/app.py
```

# Technology Stack

| Layer | Technology | Details |
|---|---|---|
| LLM | GPT-4o-mini | temperature=0, via LangChain |
| Embeddings | text-embedding-3-small | 1,536 dimensions |
| Vector DB | Pinecone Serverless | AWS us-east-1, cosine |
| Agent | LangGraph | StateGraph with conditional edges |
| UI | Streamlit | Chat interface with sidebar |
| Config | Pydantic-Settings | .env-based configuration |
| Logging | Loguru | Structured logging |
| Package mgmt | uv | Fast Python package manager |

# Limitations & Future Work

**Current limitations:**

- LLM-as-judge evaluation (no human judges yet)
- 48-question test set is relatively small
- Only K20-K21 curriculum version
- Cross-document synthesis remains weak (+0.100)

**Future directions:**

- Human evaluation study with real students
- Multi-curriculum version support (K18-K19, K22+)
- Multi-hop retrieval for cross-document questions
- Real deployment with user feedback collection
- Fine-tuned embedding model on curriculum domain

# Summary

## What We Built

An agentic RAG chatbot for curriculum advising that combines semantic vector search with deterministic structured tools, orchestrated by a LangGraph state machine.

## Key Results

- Correctness: 0.625 vs. 0.240 **(+160%)**
- Faithfulness: 0.865 vs. 0.000
- 19/48 fully correct vs. 5/48

## Key Contributions

- Hybrid retrieval architecture
- Multi-tool agentic loop
- Empirical RAG vs. baseline comparison

Thank you! Questions?